

In-Lab Programming Fundamentals

C C-112

Question 1: Count Prime Numbers in an Array

Problem Statement:

Write a C program that reads 'n' integers into an array and counts how many of them are prime numbers.

Sample Input:

Enter number of elements: 5

Enter elements: 3 4 7 10 11

Sample Output:

Number of prime elements: 3

Question 2: Frequency of Elements

Problem Statement:

Write a program in C that counts the frequency of each element in the array and displays it. Assume all elements are non-negative and less than 100.

Sample Input:

Enter number of elements: 6

Enter elements: 2 3 2 5 3 2

Sample Output:

Element 2: 3 times

Element 3: 2 times

Element 5: 1 time

Question 3: Arrange Negative Before Positive

Problem Statement:

Write a program in C that rearranges the elements of an array so that all negative numbers appear before positive numbers. The relative order among negative or positive numbers does not matter.

Constraints: Use an in-place algorithm.

Sample Input:

Enter number of elements: 8

Enter elements: 1 -2 3 -4 5 -6 7 -8

Sample Output:

Rearranged array: -2 -4 -6 -8 5 3 7 1

(Note: Order among positives or negatives may vary)